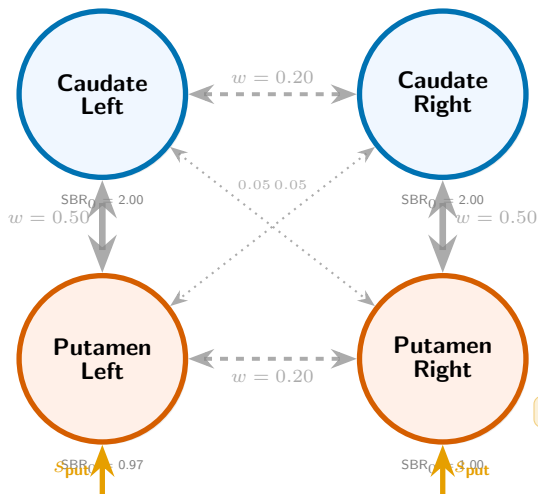


Four-Region Striatal Propagation Model



Connection types:

- Ipsilateral ($w = 0.50$)
- Commissural ($w = 0.20$)
- Contralateral ($w = 0.05$)

Parameters:

- k_{spread} : propagation rate (*fitted*)
- s_{put} : putamen seeding (*fixed*)
- k_{clear} : clearance (0.5/yr)
- β : L→N coupling (0.01)
- γ : SBR exponent (0.7)

FITTED: k_{spread}

FIXED: s_{put} , β , k_{clear}

Pathology propagation:
$$\frac{dL_i}{dt} = -k_{\text{clear}}L_i + k_{\text{spread}} \sum_j A_{ij}L_j + s_i$$

Neuron death:
$$\frac{dN_i}{dt} = -(\alpha_{\text{base}} + \beta L_i) \cdot N_i$$

Observation:
$$SBR_i(t) = SBR_{i,0} \cdot (N_i/N_{i,0})^{0.7}$$